

Problem Statement

Objectives

As a new hire to the AI Health Analytics team at ElderGuard Analytics, you have been entrusted with a critical task: predicting potential health emergencies or concerning situations for elderly residents living independently based on environmental sensor data and indoor air quality monitoring. Your goal is to help the company develop a non-invasive early warning system that can identify when elderly individuals may be experiencing distress, medical episodes, or unsafe living conditions.

ElderGuard Analytics has collected extensive data from smart home monitoring systems deployed in elderly residents' homes, including information about indoor air quality (CO2 Sensors, CO Sensor, Metal Oxide Sensors), environmental conditions (temperature, humidity), HVAC system operations, and lighting patterns throughout different data collection sessions.

By leveraging this data, you will develop predictive models that can enhance care coordination, identify patterns of concern, and provide valuable insights to family members and healthcare providers. The success of your predictions will directly impact the company's ability to provide proactive care monitoring and maintain residents' independence longer.

Specifically, your task is to build and evaluate prediction models, and also identify their respective key features of the dataset that categorise the `Activity` level of elderly residents. Your analysis should also include evaluation of which features contribute most significantly to activity level prediction.

In your submission, you are expected to build and evaluate **at least three suitable models** for this task and justify your choices based on the dataset provided.

Dataset

The dataset contains environmental sensor readings and smart home monitoring data collected from elderly residents' homes by the CareGuard Technologies monitoring systems. It includes features such as temperature, humidity, gas concentrations (CO2 Sensors, CO Sensor, Metal Oxide Sensors), HVAC operations, lighting conditions, and activity classifications across various data collection sessions.

Important Note: The dataset may contain synthetic or contaminated data. Therefore, you should state clearly any assumptions or justifications that you make in processing the data, including handling of outliers, missing values, and data quality issues.

Instructions for querying the database

The dataset can be accessed through the ``gas_monitoring.db``. You may find either of the following packages, ``SQLite`` or ``SQLAlchemy``, useful for accessing this database.

You should place the ``gas_monitoring.db`` file in a ``data`` folder. Your machine learning pipeline should retrieve the dataset using the relative path ``data/gas_monitoring.db``.

List of Attributes

Attribute	Description
Time of Day	Time period when measurements were recorded.
Temperature	Indoor air temperature measured in degrees Celsius by environmental sensors installed in the resident's living space.
Humidity	Relative humidity percentage of indoor air indicating moisture levels.
CO2_InfraredSensor	Carbon dioxide concentration readings from an infrared sensor.
CO2_ElectroChemicalSensor	Carbon dioxide concentration readings from an electrochemical sensor.
MetalOxideSensor_Unit1	Metal oxide sensor reading for detecting various gases and volatile organic compounds.
MetalOxideSensor_Unit2	Metal oxide sensor reading for detecting various gases and volatile organic compounds.
MetalOxideSensor_Unit3	Metal oxide sensor reading for detecting various gases and volatile organic compounds.
MetalOxideSensor_Unit4	Metal oxide sensor reading for detecting various gases and volatile organic compounds.
CO_GasSensor	Carbon monoxide concentration levels.
Session ID	Unique identifier for each monitoring session.
HVAC Operation Mode	Current heating, ventilation, and air conditioning system operating mode.
Ambient Light Level	Indoor lighting conditions.
Activity Level	Classified resident activity status based on environmental indicators.

